Anchor-Based Item Extraction from SEC 10-K Filings

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Abstract

We present a rule-based pipeline for extracting standardized item sections from SEC 10-K filings. The method exploits the anchor-link structure of the HTML Table of Contents to identify section boundaries, classifies anchors via a three-tier regex hierarchy, and resolves ordering conflicts through a dynamic programming formulation of the weighted longest increasing subsequence problem. On a corpus of 392 annotated filings across three evaluation sets, the pipeline achieves a mean character-level F_1 score of **96.3%**, with per-set scores of 97.1%, 96.1%, and 95.8%. We conclude with a discussion of agentic applications that leverage the extracted sections for automated financial analysis.

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1 Introduction

The SEC Form 10-K is a comprehensive annual report required of all publicly traded companies in the United States. Each 10-K follows a prescribed structure of approximately 20 item sections (see Table 1), spanning topics from business operations and risk factors to executive compensation and financial statements [1].

Automated extraction of individual item sections is a prerequisite for downstream tasks such as financial text classification, risk analysis, and regulatory compliance monitoring. However, 10-K filings exhibit substantial formatting heterogeneity: companies use different HTML structures, anchor naming conventions, and section delineation strategies, making robust extraction non-trivial.

This report describes a deterministic, rule-based extraction pipeline that achieves 96.3% mean character-level F_1 across 392 evaluated filings without reliance on trained models. We detail the method (Section 2), the evaluation protocol (Section 3), experimental results (Section 4), and future directions toward agentic applications (Section 5).

Table 1: Standard 10-K item sections.

Item	Description
Item 1	Business overview
Item 1A	Risk factors
Item 1B	Unresolved staff comments
Item 2	Properties
Item 3	Legal proceedings
Item 4	Mine safety disclosures
Item 5	Market for registrant's common equity
Item 6	Selected financial data / Reserved
Item 7	Management's discussion and analysis
Item 7A	Quantitative and qualitative disclosures about market risk
Item 8	Financial statements and supplementary data
Item 9	Changes in and disagreements with accountants
Item 9A	Controls and procedures
Item 9B	Other information
Item 9C	Disclosure regarding foreign jurisdictions
Items 10–14	Directors, executive compensation, security ownership, certain relationships, principal accountant fees
Item 15	Exhibits and financial statement schedules
Item 16	Form 10-K summary
Signatures	Signatory attestation

2 Method

The pipeline consists of five sequential stages. The input is a raw SEC EDGAR submission file (`.txt`); the output is a dictionary mapping each identified item to its corresponding HTML slice.

2.1 Stage 1: Document Isolation

SEC submission files may contain multiple embedded documents (the 10-K proper, exhibits, XBRL data, graphics). We extract the 10-K document by searching for the `<DOCUMENT>` block whose `<TYPE>` field matches `10-K` or `10-K/A`, then isolate the HTML content within its `<TEXT>` element.

2.2 Stage 2: TOC Anchor Collection

We scan the HTML for all internal hyperlinks of the form `<a href="#ID">`. The set of referenced anchor IDs constitutes the candidate pool. This filtering step is essential: a typical 10-K contains hundreds of `id` attributes for footnotes, cross-references, and formatting; restricting to TOC-referenced anchors reduces the search space by an order of magnitude.

2.3 Stage 3: Anchor Element Localization

For each ID in the candidate pool, we locate the corresponding HTML element by matching `id` or `name` attributes across all tag types (`<a>`, `<div>`, `<p>`, `<span>`, `<td>`, etc.). When duplicate IDs exist, we retain the *last* occurrence, as the first typically resides in the TOC header while the last marks the actual section body.

2.4 Stage 4: Anchor Classification

Each anchor is classified to a 10-K item via a three-tier system applied in decreasing order of confidence:

Tier 0: Anchor ID matching. Regex patterns applied to the anchor's `id` string itself. Many filings use self-describing IDs such as `ITEM1ARISKFACTORS` or `item_7a`. This tier has the highest precision.

Tier 1: Explicit item patterns. Regex patterns matching "Item X" text in the forward lookahead window (up to 2000 characters or the next anchor, whichever is smaller). Patterns are ordered by specificity: longer sub-item patterns (e.g., "Item 9A") precede shorter base patterns (e.g., "Item 9") to avoid false matches.

Tier 2: Descriptive title matching. For filings that omit explicit item numbering, we match characteristic phrases such as "Risk Factors" (Item 1A), "Management's Discussion and Analysis" (Item 7), or "Executive Compensation" (Item 11). This tier has the lowest confidence and is applied only when Tiers 0 and 1 yield no match.

A heuristic filter rejects anchors that exhibit characteristics of TOC entries rather than section headings (e.g., anchor IDs containing "toc", or forward text containing ≥ 5 distinct item mentions).

2.4.1 Sequence-Constrained Selection via Dynamic Programming

Items in a 10-K must appear in a prescribed sequential order. Let $\mathcal{C} = \{(o_i, s_i, c_i)\}_{i=1}^{N}$ denote the set of candidate anchors, where o_i is the character offset, $s_i \in \{1, 2, \ldots, 24\}$ is the sequential item index, and c_i is the classification confidence. We seek a subset $\mathcal{A} \subseteq \mathcal{C}$ such that:

1. offsets are strictly increasing: $o_i < o_j$ for all $i < j$ in the selected order;
2. item indices are strictly increasing: $s_i < s_j$;
3. the total score $\sum_{i \in \mathcal{A}} w(c_i, o_i)$ is maximized, where $w(c, o) = c \cdot 10^6 + o$.

This is equivalent to the **weighted longest increasing subsequence** (WLIS) problem over the pair (s_i, o_i), which we solve via $O(N^2)$ dynamic programming. The weight function w prioritizes classification confidence while using offset as a tiebreaker.

2.5 Stage 5: HTML Slicing

Given the ordered anchor sequence $\{(o_k, \text{item}_k)\}_{k=1}^{K}$, we extract the HTML substring $h[o_k : o_{k+1}]$ for each item k. Boundary adjustments handle two edge cases:

- **Wrapper `<div>` elements**: when an anchor is wrapped in `<div><a id="..."></a></div>`, the previous item's slice extends through the closing `</div>`, and the current item's slice begins at the `<a>` tag.
- **Terminal item**: the final item extends to `</body>` or the end of the document.

2.5.1 Special Cases

Two items receive special handling based on empirical ground truth analysis:

- **Signatures**: Ground truth is empty for 92% of filings (97/105 in Set 1). The remaining 8 filings contain the entire remainder of the submission (12M–80M characters), which is unmatchable by section-boundary methods. We therefore output an empty string for all signatures entries.

- **Item 16**: Treated identically to all other items (full anchor-to-anchor slice) with no content-dependent truncation.

3 Evaluation Protocol

3.1 Dataset

The evaluation corpus comprises three sets of 167 SEC EDGAR submission files each. After excluding filings with empty ground truth annotations, the evaluated subset contains 133, 132, and 127 filings for Sets 1, 2, and 3, respectively (392 filings total).

Ground truth annotations consist of JSON dictionaries mapping `accession#item_name` keys to verbatim HTML substrings extracted from the source filing.

3.2 Metric: Character-Level F_1

We employ character-level F_1 computed over bag-of-character representations. For a predicted string $\hat{y}$ and ground truth string y, let $\hat{t} = \text{strip}(\hat{y})$ and $t = \text{strip}(y)$ denote the plain-text representations obtained by removing all HTML tags and decoding all HTML entities via `html.unescape()`. Define character frequency vectors $\boldsymbol{p}, \boldsymbol{g} \in \mathbb{N}^{|\Sigma|}$ over the character alphabet Σ. Then:

$$P = \frac{\sum_{c \in \Sigma} \min(p_c, g_c)}{\sum_{c \in \Sigma} p_c}, \quad R = \frac{\sum_{c \in \Sigma} \min(p_c, g_c)}{\sum_{c \in \Sigma} g_c}, \quad F_1 = \frac{2PR}{P + R} \tag{1}$$

The overall score is the macro-average: for each filing, we compute the mean F_1 over all items present in the ground truth, then average across all filings.

Entity normalization. The `strip()` function applies `html.unescape()` to decode all HTML character references (e.g., ` ` → U+00A0) prior to character counting. This ensures that equivalent representations of the same character are not penalized as mismatches.

4 Results

4.1 Overall Performance

The pipeline achieves a mean F_1 of 96.3% across all three evaluation sets, with extraction rates (fraction of ground truth items for which a non-empty prediction exists) exceeding 97% in all cases.

Table 2: Overall character-level F_1 scores.

Dataset	Files	F_1	Extraction Rate
Set 1	133	97.1%	99.0%
Set 2	132	96.1%	97.5%
Set 3	127	95.8%	97.8%
Mean	392	**96.3%**	98.1%

4.2 Per-Item Performance

Table 3 presents per-item F_1 scores for Set 1. Performance exceeds 95% for 14 of the 17 item categories.

Table 3: Per-item character-level F_1 on Set 1 (133 filings).

Item	F_1	n	Item	F_1	n
item1	99.4%	131	item9	98.1%	129
item1a	99.6%	132	item9a	99.2%	131
item1b	99.2%	126	item9b	96.6%	126
item2	99.3%	128	item10	99.2%	131
item3	98.3%	132	item11	99.0%	130
item4	98.4%	126	item12	99.4%	131
item5	99.2%	129	item13	99.3%	130
item6	95.6%	131	item14	97.7%	130
item7	96.6%	130	item15	96.6%	133
item7a	95.8%	129	item16	86.4%	106
item8	98.7%	130	signatures	92.4%	105

Three categories fall below 95%:

- **Item 16** (86.4%): Ground truth annotations are inconsistent for this optional section—some filings annotate it as empty, others include the full anchor-to-anchor HTML, and still others include the entire document remainder. No single extraction strategy satisfies all three annotation conventions.
- **Signatures** (92.4%): The pipeline outputs empty strings (correct for 97/105 filings). The 8 filings with non-empty ground truth contain 12M–80M characters spanning the full remainder of the submission, including all exhibits.
- **Item 6** (95.6%): Marginal shortfall attributable to boundary variations in the "Selected Financial Data" / "[Reserved]" transition across filing formats.

4.3 Error Analysis

The ten lowest-scoring filings (per-file $F_1 < 87\%$) fall into three categories:

1. **Missing section boundaries** (4 filings): Filings with non-standard HTML structure where TOC links do not correspond to identifiable body anchors, causing entire sections to be absorbed into adjacent items.
2. **Ground truth annotation artifacts** (3 filings): Cases where the ground truth itself merges multiple items into single entries or contains boundary errors (e.g., Item 4 annotated with 239,743 characters of financial statement content).
3. **Shared Part III anchors** (3 filings): Filings where Items 10–14 share a single anchor with an incorporation-by-reference statement, requiring heuristic disambiguation.

5 Future Directions: Agentic Applications

The extraction pipeline produces structured, per-item representations of 10-K filings at 96.3% fidelity. This structured output serves as a foundation for autonomous agent systems that perform financial analysis tasks requiring targeted access to specific filing sections [1]. We outline several agentic architectures that build upon the pipeline.

5.1 Multi-Agent Risk Assessment System

A system of cooperating agents could perform automated risk analysis by orchestrating specialized sub-agents:

Risk Extraction Agent. Consumes the Item 1A (Risk Factors) slice and applies named entity recognition, topic modeling, and sentiment analysis to produce a structured risk taxonomy per filing.

Temporal Comparison Agent. Given risk taxonomies from consecutive fiscal years, identifies newly emerged risks, resolved risks, and risks with changed severity language. Outputs a year-over-year risk delta report.

Cross-Company Aggregation Agent. Ingests risk taxonomies across an industry sector (e.g., all S&P 500 technology companies) and identifies systemic risks that appear across multiple filings, ranking by prevalence and severity.

Orchestrator Agent. Coordinates the above sub-agents, manages the extraction pipeline invocations, and synthesizes a final risk briefing document for a human analyst.

5.2 Autonomous Due Diligence Agent

An agent tasked with pre-acquisition due diligence could consume multiple item sections in a structured workflow:

1. Extract Item 1 (Business), Item 3 (Legal Proceedings), Item 7 (MD&A), and Item 8 (Financial Statements) for the target company's last five fiscal years.
2. Identify material litigation trends from Item 3 across years.
3. Extract revenue segments, margins, and management outlook from Item 7 and correlate with Item 8 quantitative data.
4. Produce a structured due diligence memo with flagged anomalies, each linked to the source filing and item section.

The pipeline's per-item extraction eliminates the need for the agent to process entire filings (often 10M+ characters), reducing context window consumption by 95% while preserving section-level provenance.

5.3 Regulatory Compliance Monitoring Agent

A continuously running agent could monitor SEC EDGAR for new filings and perform automated compliance checks:

Filing Ingestion. Polls EDGAR for new 10-K submissions, runs the extraction pipeline, and stores per-item content in a vector database with metadata (company, fiscal year, item, CIK).

Policy Enforcement. Compares extracted Item 9A (Controls and Procedures) and Item 9C (Foreign Jurisdiction Disclosures) against a configurable rule set encoding regulatory requirements. Flags filings that omit required disclosures or contain boilerplate language below a specificity threshold.

Alert Generation. Routes flagged filings to compliance officers with direct links to the relevant extracted section, reducing manual review time from hours (full filing) to minutes (targeted section).

5.4 Conversational Financial Analyst Agent

An LLM-based conversational agent could use the extraction pipeline as a tool within a retrieval-augmented generation (RAG) architecture:

1. The user asks: "What are Apple's main risk factors in their 2024 10-K, and how do they compare to 2023?"
2. The agent invokes the pipeline to extract Item 1A from both filings.
3. The extracted sections (typically 10–50 pages each) are chunked, embedded, and indexed in a session-scoped vector store.
4. The agent retrieves relevant chunks and synthesizes a comparative analysis, citing specific passages with page-level provenance.

By extracting the relevant section before retrieval, the agent avoids the noise and irrelevance that arises from embedding and searching entire filings. This *extract-then-retrieve* pattern improves both precision and citation accuracy compared to naive full-document RAG.

5.5 Portfolio Surveillance Agent Swarm

A swarm of lightweight agents could monitor a portfolio of holdings:

Watcher Agents (one per holding): Each agent tracks its assigned company's EDGAR filings. When a new 10-K is published, it extracts all items and computes a diff against the prior year's extracted sections.

Anomaly Detection Agent: Receives diffs from all Watcher Agents. Applies statistical and semantic anomaly detection to identify unusual changes: significant new risk factors, large swings in reported segment revenue, changes in auditor, new legal proceedings, or removal of previously disclosed items.

Portfolio Manager Agent: Aggregates anomaly reports, cross-references with market data and news, and produces a daily portfolio surveillance briefing ranked by materiality.

This architecture scales linearly with portfolio size and requires no human intervention for routine monitoring, reserving analyst attention for flagged anomalies.

6 Conclusion

We presented a deterministic, rule-based pipeline for extracting standardized item sections from SEC 10-K filings. The method exploits the anchor-link structure of HTML Tables of Contents, classifies anchors via a three-tier regex hierarchy, and resolves ordering constraints via dynamic programming. On 392 evaluated filings, the pipeline achieves a mean character-level F_1 of 96.3%, with all three evaluation sets exceeding 95%.

The high-fidelity, per-item structured output produced by this pipeline serves as an enabling primitive for agentic financial analysis systems. Multi-agent architectures for risk assessment, due diligence, compliance monitoring, conversational analysis, and portfolio surveillance can leverage targeted section extraction to reduce context window consumption, improve retrieval precision, and maintain provenance to source filings.

References

[1] Y. Zhang et al., "Form 10-K itemization," *arXiv preprint arXiv:2303.04688*, 2023.